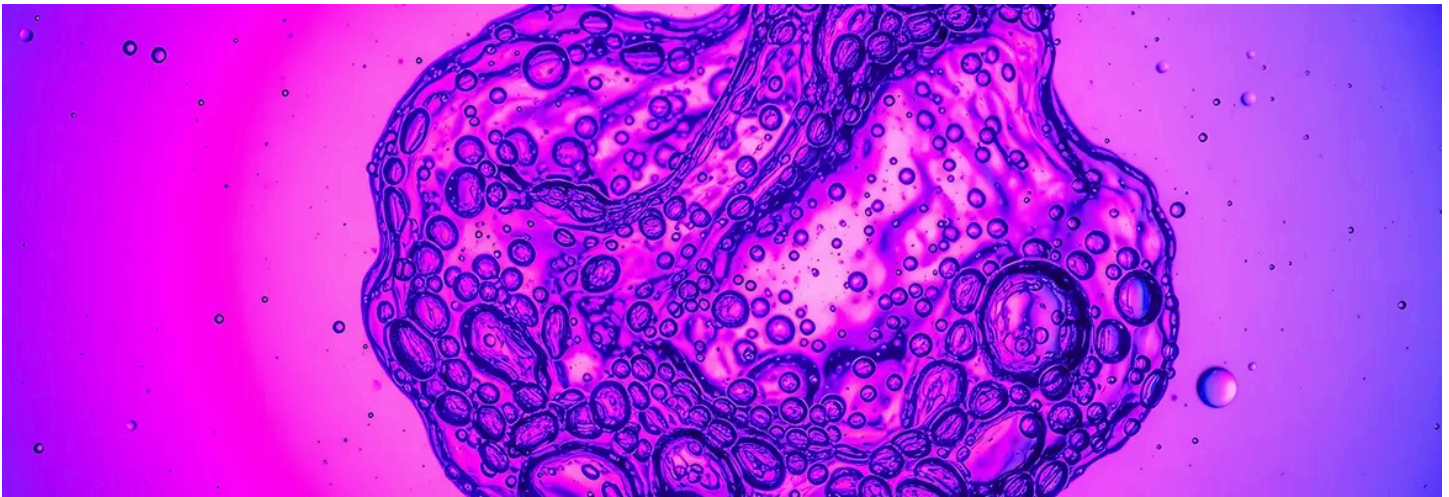


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New studies suggest unique state of matter gives rise to living organisms

Organisms are processes, not things



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Addy Pross | Emeritus Professor of Chemistry at Ben Gurion University of the Negev.

1,437 words

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*How can inert, passive matter give rise to life, with all its agency and purpose? Chemist **Addy Pross** has spent 25 years on this mystery, culminating in a landmark book, *Life's Chemical Secret: The Origin of Purpose and Mind*. Now, he says, fresh evidence backs up his theory of how matter comes to life. We know matter can be stable when collected into an object which settles into a low-energy, resting state, like a rock at the bottom of a hill. But matter can also have a different sort of stability, as part of a process which persists through constant exchanges of material with its surroundings, like a flowing river. Life, Pross thinks, emerges from such states of constant interaction. Indeed, he argues here, new experiments suggest that as these interactions grow more complex, matter behaves in ways that look increasingly purposeful.*

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For four centuries, science has been based on the fundamental assumption that nature is, at its core, indifferent. Rocks, rivers, stars—they are all without desires and intentions. From atoms and molecules up, matter is nothing more than physical systems obeying physical laws. As French biologist Jacques Monod famously put it in his classic 1970 work *Chance and Necessity*: nature is objective. That idea encapsulated the essence of the modern scientific revolution, as forged by the landmark thinking of Galileo, Descartes, Newton and others. And thanks to Darwin's theory of evolution, biology seemed to also fit into that physicalist framework, at least at first.

Yet, with regard to life, one difficulty has refused to go away. Living things **express characteristics that are inconsistent with that physical/mechanistic view**: purpose, desire, intentionality, subjectivity are all undeniable aspects of reality. But how can such characteristics fit in with that mechanistic view? **The riddle of how subjective systems were able to emerge from an objective universe** remained troublingly unresolved.

But, as I will now describe, **an answer may be within reach**, and it lies in the recent discovery that a certain category of matter can be actively creative in ways that the classical picture overlooked.

The overlooked state of matter

We are all familiar with the three classical states of matter—solid, liquid, gas—with their distinctly different physical properties. A crystal of ice, a drop of water, and a puff of steam are all composed of the same H₂O molecules, but their different arrangements give rise to those different properties.

But what has not been appreciated till recently is that there exists a fundamentally different material arrangement of matter, one defined not by equilibrium structures, but by non-equilibrium dynamic processes, one that was only identified relatively recently. It is within that dynamic domain that matter's subjective and **creative potential** can be unearthed.



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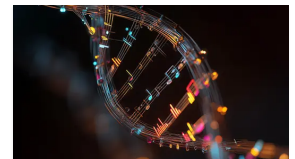
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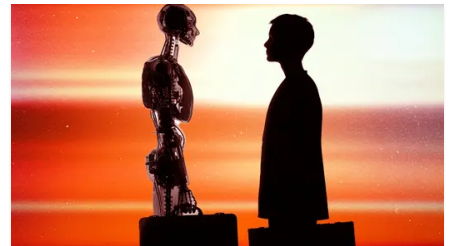
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